

Semester project Report: Crop Classification from Time-Series Satellite Imagery

Jeffrey Zweidler
Computer Science Master
ETH

Dr. Mehmet Ozgur Turkoglu
Earth Observation of Agroecosystems
Agroscope

Dr. Helge Aasen
Earth Observation of Agroecosystems
Agroscope

Prof. Konrad Schindler
Photogrammetry and Remote Sensing
ETH

Abstract—This project applies the ViTs for SITS (Vision Transformers for Satellite Image Time Series) model to Swiss Sentinel satellite data, focusing on mid-season crop classification. While the model performs well on the original PASTIS benchmark, applying it to Swiss data revealed significant challenges due to domain shift. In particular, smaller and more heterogeneous field patterns, as well as differing crop distributions, complicated model learning. The project also explores the integration of temperature measurements as a way to include environmental variables relevant to crop growth. Results highlight the need for region-specific adaptations when transferring remote sensing models across domains.

I. INTRODUCTION

In this project, we address the task of pixel-wise segmentation of crop types using Sentinel-2 satellite image time series. The goal is to classify each pixel in a region into one of several crop classes based on its temporal signature over a growing season. This approach has significant potential for agricultural monitoring, as it enables large-scale, automated crop mapping with high spatial and temporal resolution. For model architecture and methodology, we build on the work presented in ViTs for SITS: Vision Transformers for Satellite Image Time Series [1], which introduced a novel adaptation of the Vision Transformer (ViT) model for processing satellite time series data. The core idea of the model is to treat time as a sequence of image tokens, using transformer-based self-attention to capture both spatial and temporal dependencies across the sequence. This enables the model to learn complex crop phenology patterns directly from raw satellite observations.

A promising new approach suggests the integration of environmental variables into the purely vision based time series [2]. In particular, this research demonstrates that replacing simple day-of-year (DOY) timestamps with temperature-based time encoding improves model performance. This suggests that embedding physiologically relevant environmental information can help the model better align observations across seasons and regions, and learn more robust temporal representations. This work tests the importance of environmental variables in crop classification using a Vision Transformer as basis.

The main focus of this project is the adaptation of the ViTs for SITS model to Swiss Sentinel-2 data. Unlike the benchmark datasets used in the original work, Switzerland presents unique challenges due to its alpine regions which comprise of smaller, more heterogeneous agricultural fields and different crop distributions. We investigate how this domain shift affects model performance and what adaptations may be necessary for successful transfer. Additionally, we evaluate the feasibility of mid-season classification — identifying crops before harvest — which holds practical relevance for public sector applications. These include early yield estimations, monitoring compliance with subsidy and land use programs, and providing region-specific recommendations for fertilization and land management practices. Through this work, we aim to assess the applicability of state-of-the-art remote sensing models to real-world use cases in diverse agricultural settings.

II. DATASET

The PASTIS dataset was introduced as a benchmark for semantic segmentation of satellite image time series, specifically designed to evaluate temporal deep learning models for crop mapping. It covers a limited agricultural region in Southwest France and provides high-quality ground truth annotations for 19 crop classes, making it a compact yet challenging testbed for methodological development. Data splits are created by partitioning patches from the same year into training, validation, and test folds. The CropMapsSwitzerland dataset serves as an analogous benchmark for the Swiss context, but at a national scale. Instead of a restricted region, it includes all agricultural areas across Switzerland, offering a comprehensive view of the country’s diverse cropping systems. With 53 crop and land cover classes, it provides a more fine-grained and heterogeneous setting compared to PASTIS. In contrast to PASTIS, data splits are based on different years of imagery, which adds temporal variability to the evaluation. A compact overview is given in Tab. I

TABLE I: PASTIS vs CropMapsSwitzerland

Dataset	Year(s)	# Crops	Total Area km ²
PASTIS	2019	19	3'114
CropMapsSwitzerland	2021-2023	53	78'566

Short fact sheet on both datasets

A. Label Taxonomies: PASTIS vs CropMapsSwitzerland

A central difference between the benchmark dataset used in ViTs for SITS — PASTIS — and the CropMapsSwitzerland dataset used in this project lies in their class granularity and handling of minor categories. PASTIS defines 19 crop classes and includes a relatively large “void” class that absorbs rare crops and ambiguous land cover types not explicitly annotated. This reduces class diversity and simplifies the classification task by filtering out many infrequent or noisy categories. Beyond imbalance, the CropMapsSwitzerland label taxonomy is also more fine-grained than in PASTIS. Where PASTIS groups various crops into broad categories — such as its “fruits, vegetables, and flowers” class — CropMapsSwitzerland separates these into individual crop types. In total, CropMapsSwitzerland defines 53 distinct classes, covering both crops and non-agricultural land covers such as forests, water bodies, roads, and hedgerows. While it also includes a void category, it is comparatively small, since most minority crop classes and heterogeneous land covers are retained explicitly. These differences are reflected in the pixel distribution plots for both datasets (Fig. 1). PASTIS exhibits a relatively compact class distribution, with a heavy tail masked by the void class. CropMapsSwitzerland, by comparison, reveals a long-tailed distribution across all classes, introducing a high degree of label imbalance. The contrast is further illustrated in side-by-side visual examples of labeled image patches from both datasets (Fig. 2a vs Fig. 2b): PASTIS patches appear more homogeneous and simplified, while CropMapsSwitzerland patches show dense spatial variation and more detailed semantic structure.

B. Dataset size and format

Working with the CropMapsSwitzerland dataset introduced significant technical challenges due to its scale and storage format. The dataset consists of three years of Sentinel-2 observations, covering all agricultural areas across Switzerland. Each year amounts to approximately 1.5TB of data, stored in the standard .zarr format commonly used for Sentinel products. While .zarr is efficient for cloud-based storage and lazy loading, it results in an extremely high file count — a factor that quickly became a limiting constraint on the CSCS HPC cluster used for this project. File count restrictions are a known bottleneck on modern HPC systems, and in this case, the original data pipeline provided by Agroscope led to severe I/O congestion. This significantly slowed training, making experimentation impractically time-consuming.

To address this, I explored alternative data storage

solutions and ultimately restructured the dataset using the WebDataset format. While promising in theory, adapting to WebDataset proved more difficult than expected due to sparse documentation and a number of stability issues within the package. Several core features were either poorly documented or non-functional, requiring custom implementations to replace or extend the intended functionality. Despite these setbacks, the transition to WebDataset resulted in a substantial improvement in data loading speed (Tab. II & Tab. III). A comparison of sample and batch load times before and after the transition illustrates the effectiveness of this change. With this new setup, I was able to bypass the file count limitations of the HPC system and remove the primary bottleneck in the training pipeline.

TABLE II: Single Sample Load Time Comparison

Component	SquashFS	Webdataset
Load Time	0.0462s	0.0002s
Transforms	0.0138s	0.0002s

Times are averaged over 200 samples.

TABLE III: Batch Load Time Comparison

Num workers	SquashFS	Webdataset
1	1.7402s	0.0978s
4	1.3599s	0.0295s
8	1.9753s	0.0179s

Times are averaged over 200 batches. Batch size of 64 samples.

III. METHODS

A. Model specifications

In this section, I provide a brief overview of the ViTs for SITS model architecture, focusing only on the aspects most relevant to this project. I do not cover the full set of architectural or training details, pointing towards the original paper [1]. The model is specifically designed for pixel-wise classification of satellite time series data and consists of two main components: a temporal transformer and a spatial transformer. The temporal transformer first encodes the evolution of each pixel (or patch) over time, modeling phenological patterns across the growing season. This output is then passed into the spatial transformer, which captures contextual information from neighboring patches within a single timestamp. The final encoded representation is fed into a classification head that predicts the crop class for each pixel.

One design choice with significant implications for this project is the use of class tokens (CLS tokens). Instead of using a single global CLS token as in standard ViTs, the authors introduce one CLS token per class. For a dataset with C classes, this results in C CLS tokens, each capturing class-specific features. While this design can enhance class discrimination, it introduces drawbacks in terms of memory and training stability. The attention mechanism scales quadratically with the number of tokens, so adding multiple CLS tokens increases

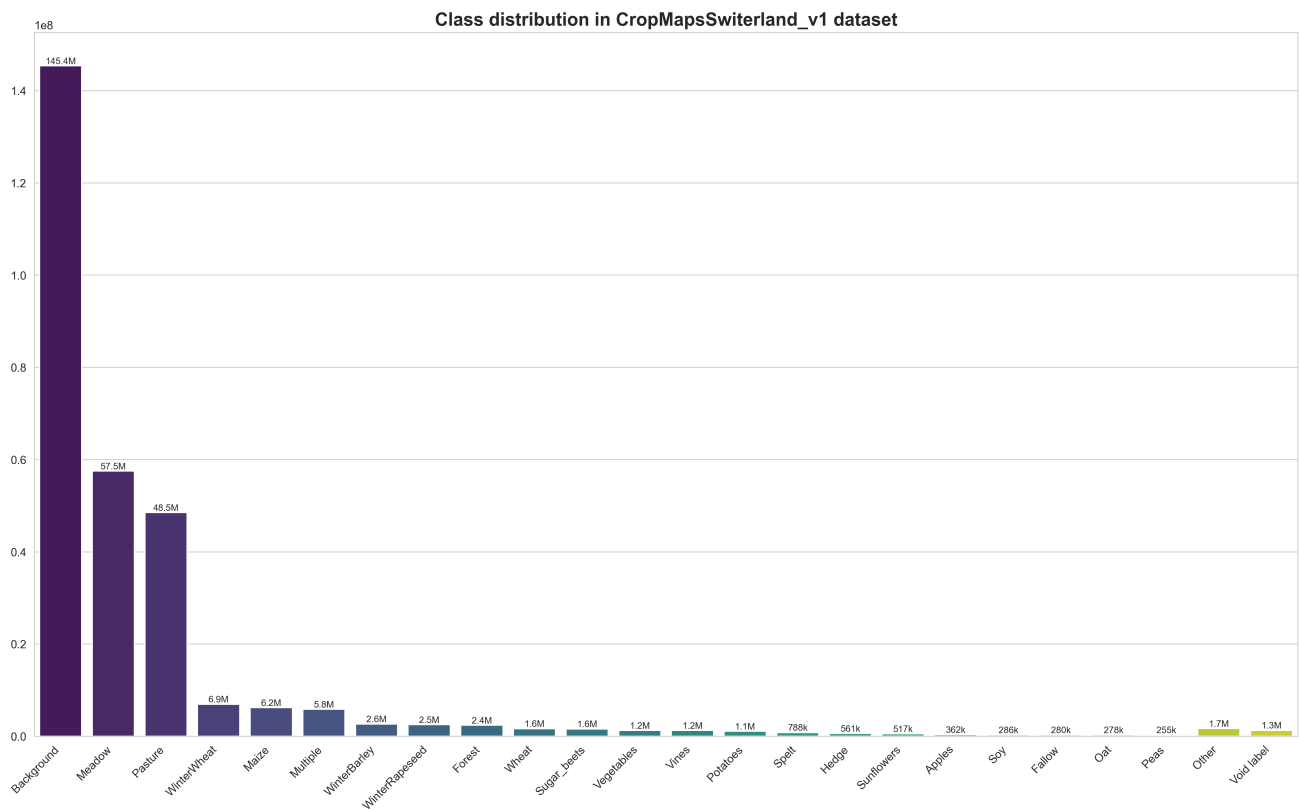
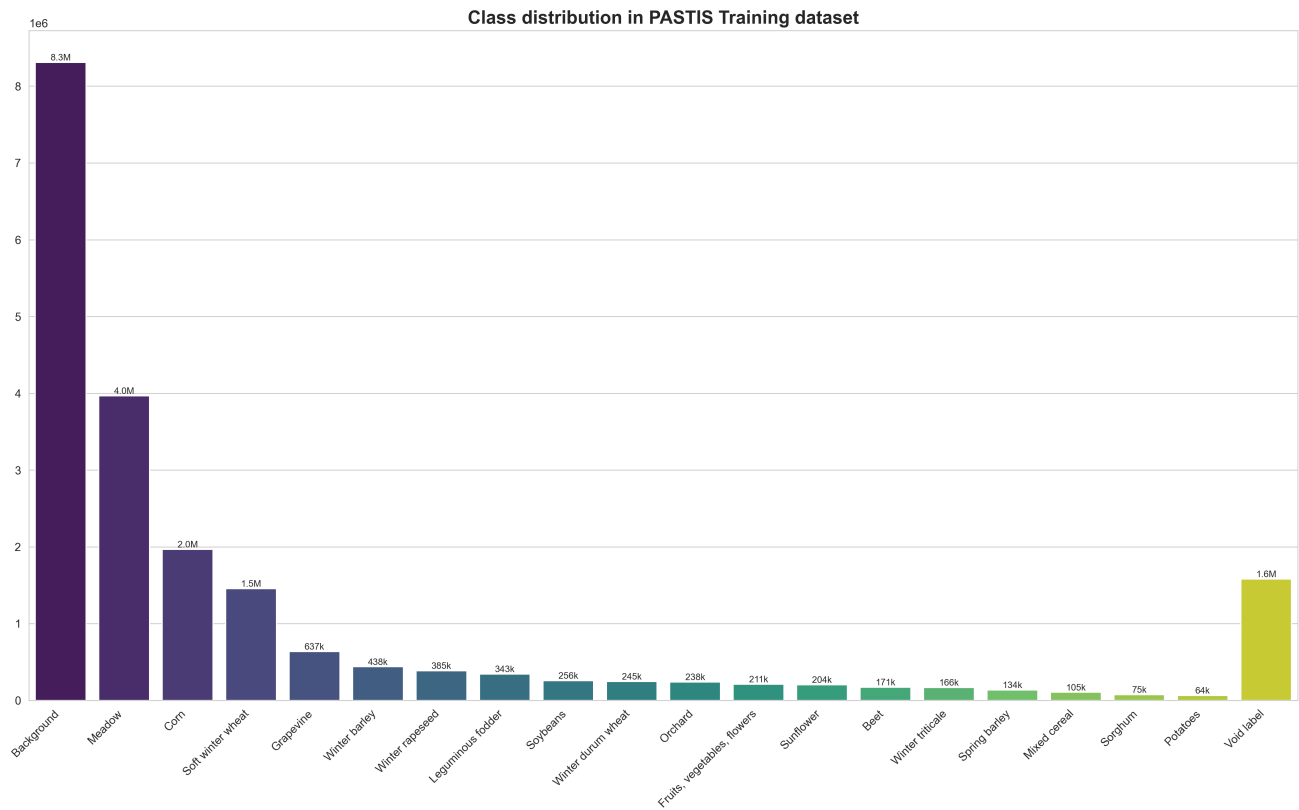
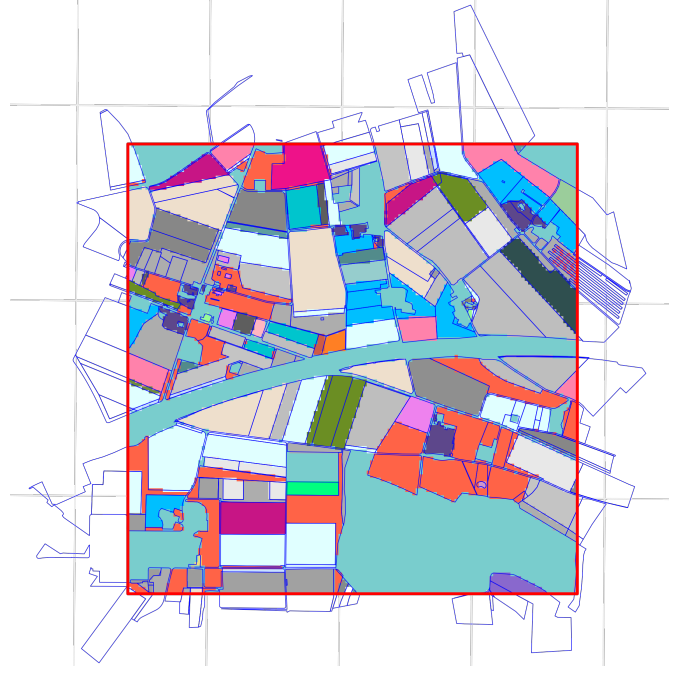


Fig. 1: Class distributions for (top) PASTIS and (bottom) CropMapsSwitzerland.



(a) Annotated crop parcel map: PASTIS



(b) Annotated crop parcel map: CropMapsSwitzerland

Fig. 2: Comparison of crop distributions.

memory usage substantially. Furthermore, the diffusion of softmax values across many tokens can lead to noisy gradient updates, especially when class distributions are imbalanced. These effects are particularly relevant in our setting, where the CropMapsSwitzerland dataset includes 53 classes. Fig. 3 from the original paper illustrates this CLS token setup.

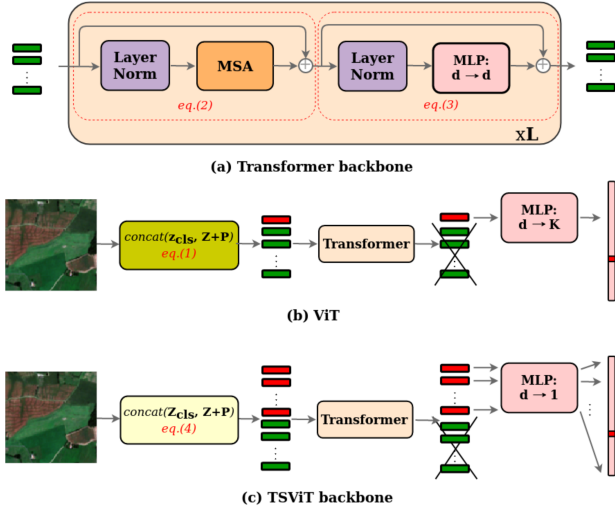


Fig. 3: CLS tokens used in TSViT

Another architectural detail relevant to our work is the tokenization strategy (Fig. 4). The model uses a patch size of 2x2 pixels, which is already quite small by ViT standards. However, in heterogeneous landscapes like those represented in CropMapsSwitzerland, even this resolution can be too

coarse. Fine-grained spatial detail is often lost, especially at class boundaries or in regions with small, fragmented field structures. This issue was less apparent in the original PASTIS dataset due to its larger, more homogeneous fields. The effect is particularly noticeable when using the original model’s lightweight, single-layer MLP decoder head. While this design choice keeps the architecture simple and efficient, it may be insufficient for accurately resolving the finer distinctions required in more complex settings like CropMapsSwitzerland.

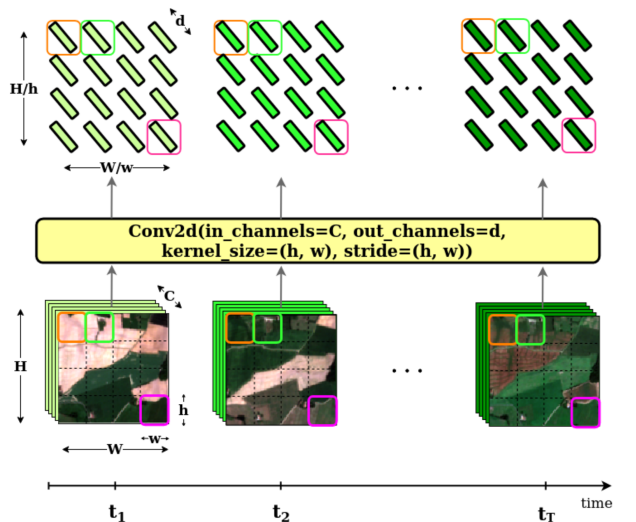


Fig. 4: Patching in TSViT

B. Loss Functions and Evaluation Metrics

Cross-Entropy Loss (CE) Cross-entropy loss is a standard objective for multi-class classification. For a single sample with true class y and predicted probability distribution $\mathbf{p}$, it is defined as:

$$\mathcal{L}_{\text{CE}} = - \sum_{c=1}^C y_c \log(p_c)$$

where C is the number of classes, y_c is a one-hot encoded label, and p_c is the predicted probability for class c .

Focal Loss Focal loss modifies cross-entropy to address class imbalance by down-weighting easy examples and focusing learning on hard, misclassified samples:

$$\mathcal{L}_{\text{Focal}} = - \sum_{c=1}^C \alpha_c (1 - p_c)^\gamma y_c \log(p_c)$$

where α_c is a weighting factor for class c , and γ controls the strength of focusing on hard examples.

Evaluation Metrics Let TP_c , FP_c , and FN_c be the true positives, false positives, and false negatives for class c , respectively.

Micro-averaged metrics compute global counts over all classes before calculating the score:

$$\text{Acc}_{\text{micro}} = \frac{\sum_{c=1}^C TP_c}{\sum_{c=1}^C (TP_c + FP_c)}$$

$$\text{IoU}_{\text{micro}} = \frac{\sum_{c=1}^C TP_c}{\sum_{c=1}^C (TP_c + FP_c + FN_c)}$$

In contrast, **macro-averaged metrics** first compute the score per class and then take the average over classes, treating all classes equally regardless of their frequency. This means micro metrics are dominated by large, frequent classes, while macro metrics give equal weight to rare classes.

IV. IMPLEMENTATION

A. Reproducing PASTIS Benchmark Results

As a first step, I aimed to reproduce the results from ViTs for SITS on the PASTIS dataset. This process turned out to be considerably more involved than expected due to multiple issues in the provided codebase. Several parts of the original training script were broken, and the project relied on outdated dependencies and Linux-specific commands. Adapting the code to run in a modern environment required resolving major package compatibility issues and replacing unsupported functionality. Once the code was operational, I encountered significant training instabilities. In practice, only about one out of five runs converged to a reasonable solution, while the others exhibited erratic behaviour, as shown in Fig. 5. Tracking the training dynamics revealed that exploding gradients were the underlying cause. My initial attempt to resolve this via gradient clipping had mixed results: clipping values that were too small or too large caused the model to fail to converge entirely.

To address this, I instead applied attention residual truncation within the transformer blocks, which provided stable training across runs without the sensitivity issues of gradient clipping (Fig. 6). With this modification in place, I was able to consistently reproduce the results reported in the original paper (Tab. IV).

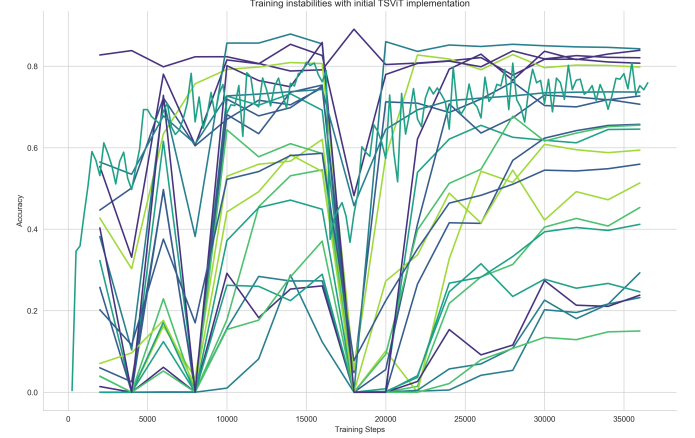


Fig. 5: Initial training instabilities with the original TSViT

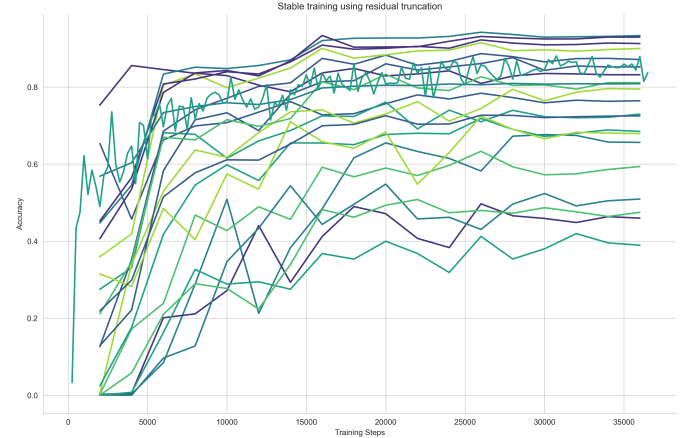


Fig. 6: Stable training via residual truncation in the TSViT

B. Adaptation to CropMapsSwitzerland

Applying the exact same training setup from the PASTIS experiments to the CropMapsSwitzerland dataset yielded the results shown Tab. IV. While the mean accuracy and IoU remained in an acceptable range, the macro accuracy and macro IoU dropped sharply. A breakdown of class-wise metrics revealed a strong bias toward the largest classes — specifically background, meadows, and pastures. Together, these three classes account for roughly 73% of all pixels in the dataset, which explains why the model focused disproportionately on them. This behaviour is a well-known drawback of using standard cross-entropy loss without class rebalancing in highly imbalanced datasets. Cross-entropy implicitly weights each sample equally, meaning classes with more pixels dominate the

loss signal. As a result, the model can minimize its overall loss simply by improving predictions on large classes, while largely ignoring smaller ones. This leads to high mean performance metrics but poor macro-level performance, where all classes are weighted equally in evaluation.

TABLE IV: Unweighted Cross Entropy PASTIS vs CropMapsSwitzerland

Metrics	PASTIS		CropMapsSwitzerland	
	Accuracy	IOU	Accuracy	IOU
Micro	0.808	0.678	0.760	0.613
Macro	0.725	0.601	0.350	0.277

Reproduced results on PASTIS vs Initial results on CropMapsSwitzerland

To address this, I experimented with class-weighted cross-entropy, aiming to give more influence to underrepresented classes in the loss function. In theory, this should encourage the model to pay greater attention to minority classes. However, in practice the results were unstable. When the weights were set too low, they had no meaningful impact on the training outcome. Conversely, when the weights were set too high, training became unstable, as shown in Fig. 7. This instability can be explained by the fact that large class weights effectively amplify the loss for certain samples, which in turn scales up their gradients. Extremely large gradients can overwhelm the optimizer’s update step, causing parameters to change too drastically from one iteration to the next. In transformers, this can cascade through multiple layers, leading to erratic attention patterns and divergence rather than convergence.

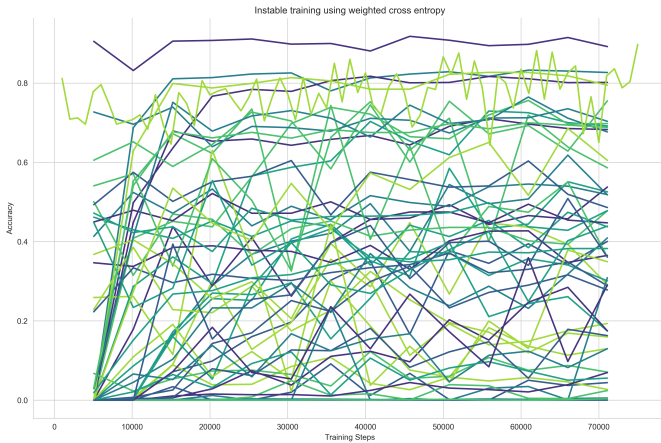


Fig. 7: Unstable weighted Cross Entropy Training on CropMapsSwitzerland

C. Empty patch detection and removal

The observations from the CropMapsSwitzerland adaptation experiments led me to take a closer look at the dataset composition. While the original 128×128 input images contained no completely empty images (i.e., all background), the

model processes smaller 16×16 patches. At this resolution, a substantial number of patches contained only background. Specifically:

- Total patches: 1,319,360
- Empty patches: 424,360

Removing these empty patches from the training set would significantly reduce the dominance of the background class. However, in a real-world deployment scenario, we would not have access to ground truth labels to identify empty patches in advance. To address this, I trained the same ViTs for SITS model in a binary classification setting to distinguish between background and foreground (all other classes combined). The classifier outputs a sigmoid score, and by adjusting the classification threshold, we can control the trade-off between removing empty patches and avoiding false negatives (patches incorrectly predicted as empty). In this context, minimizing false negatives is critical, as each one represents a patch with meaningful content that would be incorrectly excluded from training.

The evaluation results are shown in Fig. 8. The horizontal dashed line marks the total number of empty patches. The predicted empty patches decrease as the threshold increases, while the false negatives also drop. At an optimal threshold, it is possible to remove nearly 400,000 patches while introducing only about 80,000 false negatives. These false negatives are likely to be sparse patches containing very little useful information, so their loss is unlikely to harm model performance significantly.

It is worth noting that this binary classification experiment was not exhaustively tuned. Improved results could likely be achieved through careful hyperparameter selection, alternative loss functions, or specialized model architectures. Nevertheless, these findings demonstrate that omitting empty patches is a viable way to reduce dataset complexity and background class dominance at a manageable cost.

D. Class Rebalancing

With the dataset now simplified through empty-patch removal, I was able to apply class weighting without running into the severe training instabilities encountered earlier. I tested both Cross Entropy (CE) and Focal Loss in weighted and unweighted configurations, where Focal Loss consistently achieved better results.

The unweighted baseline achieves decent macro precision but suffers from poor macro recall. This behavior is expected in imbalanced datasets: when using Focal Loss without rebalancing, the loss is dominated by the most common classes.

Increasing the class weighting parameter improves recall substantially — because the model is now penalized more for missing small-class pixels — but comes at the cost of a sharp drop in precision (Fig. 9). This happens because the stronger

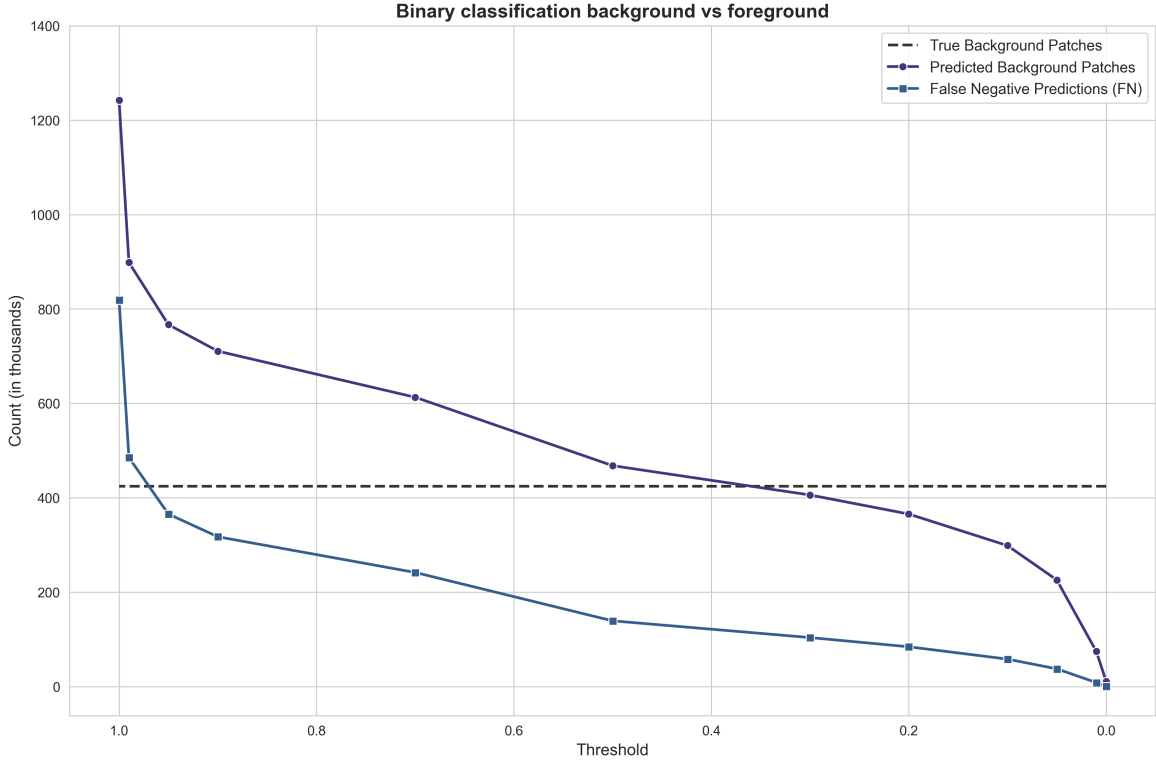


Fig. 8: Binary classification on background at different thresholds

push towards small classes also increases false positive count (likelihood of misclassifying large-class pixels as smaller classes), leading to lower precision and, in our case, a net decrease in IoU .

From these results, one can conclude that the best overall IoU is still obtained with small to none class rebalancing — at the cost of overfitting to large classes. This effect is visible in the targeted confusion matrix below (Fig. 10), which focuses on a subset of minority classes alongside a few dominant classes to visualize this effect. It shows a common failure mode: when uncertain, the model defaults to predicting one of the large classes. Class rebalancing alleviates this by improving recall, but the accompanying precision loss often outweighs the gain in IoU. For our final configuration we settled for slight class rebalancing ($\alpha = 0.1$), as IoU is steady while we trade off a little precision for some recall.

E. Alternative Loss functions

I also experimented with more sophisticated loss functions, including ones designed to:

- Directly optimize IoU.
- Allow explicit tuning of the false-positive vs. false-negative trade-off.

These “fancier” losses were significantly harder to train. This is because IoU-based objectives are non-linear and non-decomposable across pixels, so their gradients can be noisy and unstable — especially when dealing with rare classes that contribute little signal in each batch. Moreover, loss functions

with explicit FP/FN trade-off parameters can be very sensitive to hyperparameter choice, requiring careful tuning of both learning rates and class weights. Their full potential remains unknown in our setting, as a more exhaustive search was beyond the scope of this project. Fig. V below shows initial results from these experiments.

TABLE V: Alternative loss function metrics

Loss function	Metrics	Scores	
		Accuracy	IOU
Log Cosh Dice	Micro	0.687	0.523
	Macro	0.353	0.284
Lovasz-Softmax	Micro	0.691	0.528
	Macro	0.430	0.295
Focal Tversky (instable)	Micro	0.493	0.327
	Macro	0.144	0.074

Alternative loss functions were unable to improve results over standard losses given the limited time for hyperparameter tuning.

F. Model Adaptations for CropMapsSwitzerland

To better adapt the transformer architecture to the more diverse and challenging setting of CropMapsSwitzerland, two modifications were explored. The first approach was to increase the overall model size and parameter count. Since CropMapsSwitzerland defines 53 crop and land cover classes compared to the 19 classes in PASTIS, the model requires 53 classification tokens, potentially introducing a bottleneck in

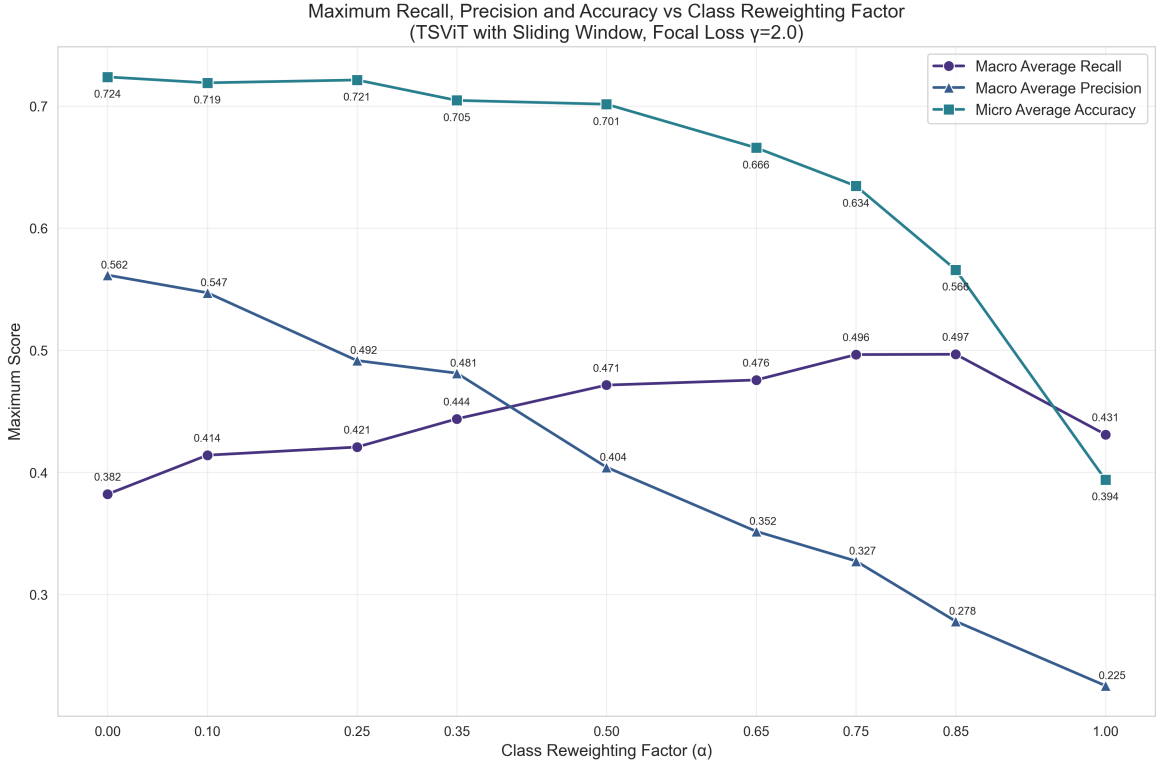


Fig. 9: Recall vs Precision tradeoff when applying class weights

representational capacity. However, doubling the model size did not yield any improvement in evaluation performance, suggesting that model capacity was not the limiting factor.

A second approach targeted the heterogeneous nature of Swiss agricultural fields. Instead of classifying aggregated 2×2 patches, the patch size was reduced to 1, allowing each pixel to be classified individually through the transformer pipeline. This aimed to capture finer spatial detail and reduce information loss in mixed-field settings. Nevertheless, this modification also showed no measurable effect on performance, indicating that architectural adjustments alone may not be sufficient to address the challenges posed by CropMapsSwitzerland. Therefore I proceeded with the standard model of TSViT and the insights in hyperparameter and loss function calibration touched upon in the previous sections.

V. EXPERIMENTS & RESULTS

A. Integration of temperature data

The first experiment tested whether replacing the temporal encoding (timestamps) with aggregated temperature data would improve model performance. This idea was inspired by an unpublished paper by [Author name], which reported substantial gains using aggregated temperature as a temporal feature instead of raw timestamps.

In my setting, the change made no meaningful difference.

The performance metrics, shown in Tab. VI, varied only within the typical margin of error expected from different random initializations, with a slightly bias towards majority classes. This indicates that, for this specific model and dataset, integrating environmental data neither improves nor harms learning in a statistically significant way.

TABLE VI: Temporal encoding: Day-of-year vs Aggregated Temperature

Metrics	Doy		Temperature	
	Accuracy	IOU	Accuracy	IOU
Micro	0.719	0.562	0.731	0.576
Macro	0.399	0.308	0.379	0.286

Aggregated temperature encodings provided a slight bias towards majority classes in comparison to Day-of-year encodings, similar to non-weighted segmentation.

B. Whole year classification results

To evaluate how temporal variation in the dataset affects performance, I conducted experiments using three years of data (2021, 2022, 2023). In each fold, one year is used exclusively for training, one for evaluation, and the remaining year for testing. This ensures that every year is used once as the training set while the other two serve as unseen validation and test data. The exact fold configurations are given in Tab. VII.

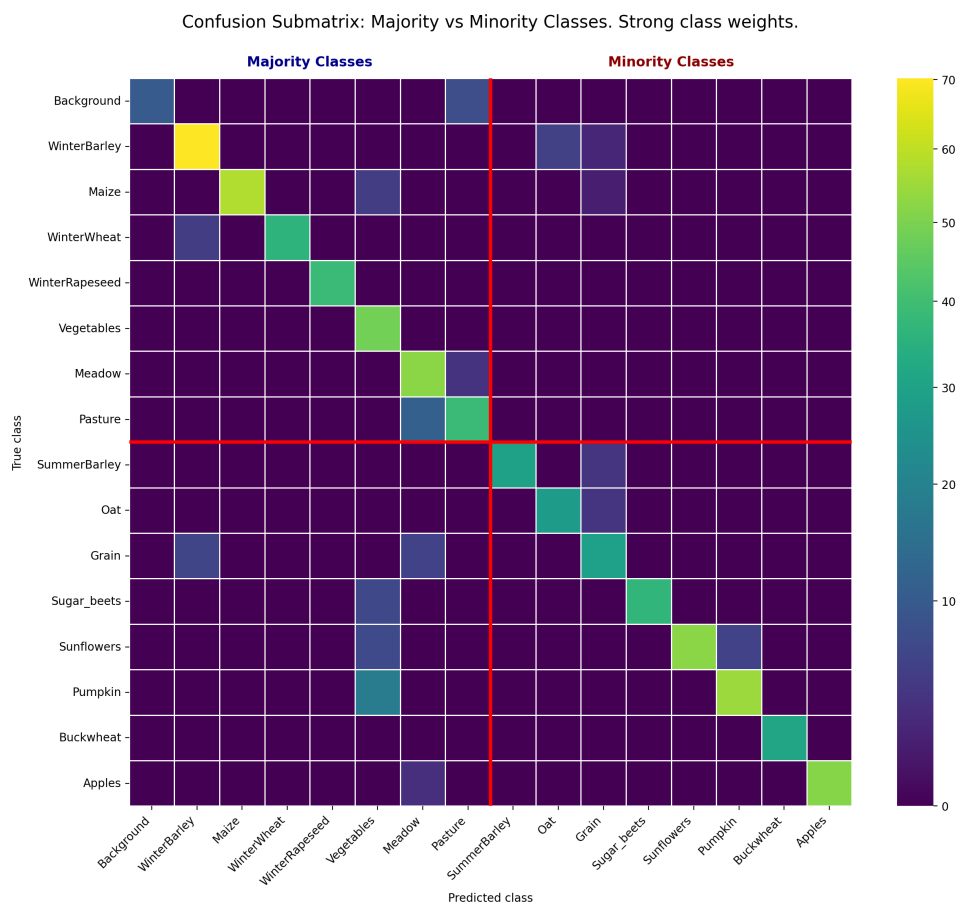
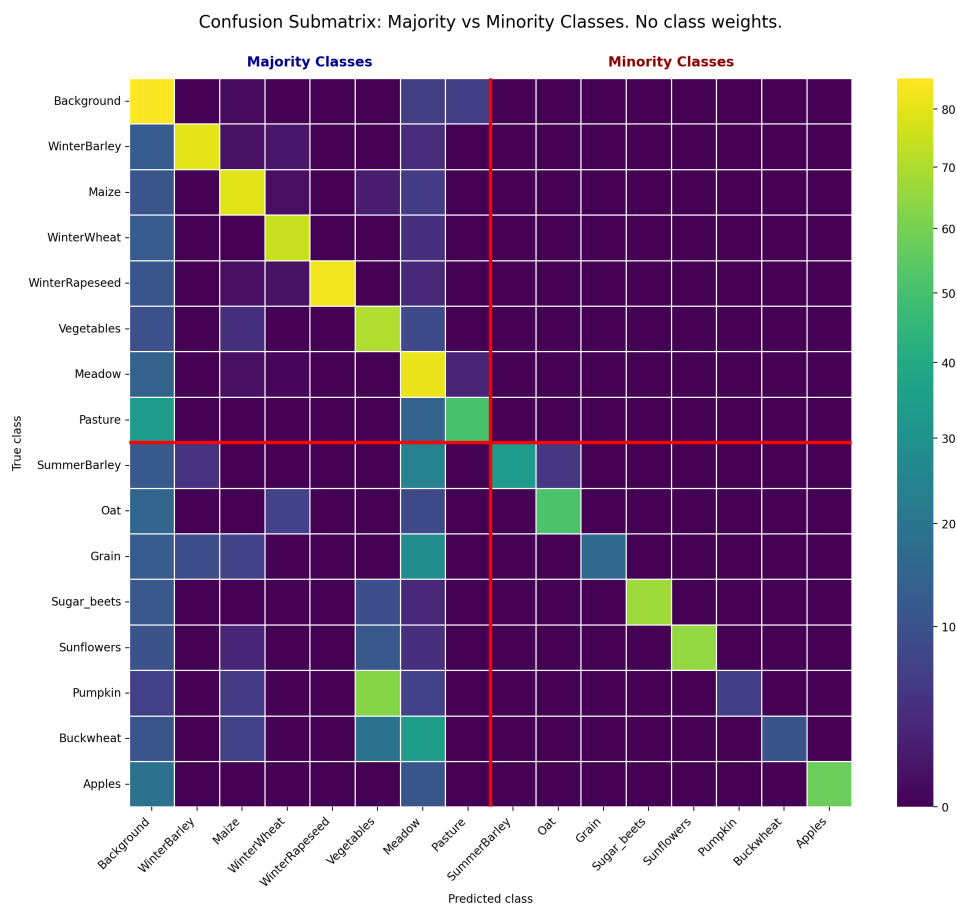


Fig. 10: Confusion matrix for a subset of the 53 classes showcasing selected majority and minority classes. No class rebalancing (top) vs strong class rebalancing (bottom).

TABLE VII: CropMapsSwitzerland folds

Fold	Train	Eval	Test
1	2022	2021	2023
2	2023	2021	2022
3	2021	2022	2023

Fold configurations

Across all folds, accuracy and IoU remain in a similar range, suggesting that the model generalizes reasonably well between years (Fig. VIII). The best performance is achieved when training on the 2023 data. This is likely due to 2023 containing consistently higher-quality imagery, with fewer cloudy acquisition days compared to the other years. Higher image clarity likely provides the model with cleaner, more representative training signals, leading to improved generalization.

TABLE VIII: CropMapsSwitzerland metrics

Folds	Metrics	Eval		Test	
		Accuracy	IOU	Accuracy	IOU
1	Micro	0.721	0.564	0.692	0.529
	Macro	0.397	0.307	0.385	0.285
2	Micro	0.721	0.564	0.726	0.570
	Macro	0.383	0.284	0.444	0.331
3	Micro	0.702	0.541	0.718	0.560
	Macro	0.366	0.267	0.376	0.283

Inter-year performance metrics for CropMapsSwitzerland

C. Midseason classification results

Fig. 11 shows midseason classification performance (Accuracy, IoU) for a selection of crop types when the temporal input is limited to data up to the end of April, May, and June. The chosen crops represent classes that typically exhibit sufficient phenological development to be discriminated by remote sensing within this period. All other crops have been masked in this setting. Overall, performance improves consistently from April to June across most crop types, reflecting the increasing spectral separability of crops as they progress through their growth stages. For example, soft winter wheat and winter barley already show relatively strong performance in April, which can be explained by their early growth cycle in Switzerland — both establish during winter and enter rapid vegetative development in spring, making them spectrally distinct earlier in the season. In contrast, maize (corn) shows very low performance in April and May but improves markedly by June, which aligns with its later sowing period and subsequent emergence and canopy development visible only from early summer onwards. Similarly, crops like potatoes or sugar beet achieve modest accuracy in April but improve by June, reflecting their gradual establishment during late spring. This temporal trend highlights that classification quality is strongly tied to crop phenology: early-sown winter crops become separable much sooner, while spring-sown crops require later observations before they can be reliably distinguished.

D. Class aggregations in CropMapsSwitzerland

To enable a more direct comparison between the PASTIS and CropMapsSwitzerland datasets, I applied class aggregation to the CropMapsSwitzerland labels. The goal was to reduce the total number of classes to a level closer to PASTIS, thereby addressing two key issues observed earlier:

- 1) Gradient dilution from excessive CLS tokens – In ViTs for SITS, the number of CLS tokens scales with the number of classes. A large class count increases memory usage and can weaken gradient signals for smaller classes.
- 2) Severe class imbalance – Reducing the class set helps bring the pixel distribution closer to uniform, mitigating the dominance of a few large classes.

The aggregation was performed based on two criteria:

- Closeness to the PASTIS class definitions
- Visual and seasonal similarities between crops to ensure that the merged classes remain meaningful in an agricultural context

A visualization of the aggregation mapping is provided in Fig. IX. While this adjustment aligns class counts and reduces imbalance, it does not address the structural difference between datasets: the more heterogeneous and fragmented fields in CropMapsSwitzerland compared to the more homogeneous fields in PASTIS.

With this class aggregation applied, model performance on CropMapsSwitzerland improved significantly, matching the scores better achieved on PASTIS (Fig. X). This creates an opportunity for hierarchical classification approaches in future work, where coarse aggregated predictions could serve as a foundation for refining into more detailed class distinctions.

VI. CONCLUSION

This work investigated the adaptation of the ViTs for SITS transformer-based model for crop type mapping, moving from the well-established PASTIS benchmark to the more diverse and challenging CropMapsSwitzerland dataset. While PASTIS offers a relatively controlled setting with homogeneous fields and a limited set of 19 classes, CropMapsSwitzerland introduces several additional difficulties: a substantially larger class set of 53 categories (covering both agricultural and non-agricultural land), stronger class imbalance, and far more heterogeneous field structures.

Our experiments confirmed that these dataset-level differences strongly influence model behavior. On CropMapsSwitzerland, micro-averaged metrics remain close to those achieved on PASTIS, but macro-averaged performance lags behind significantly. This reflects the difficulty of correctly classifying minority classes and crops occurring in fragmented or spectrally ambiguous fields.

To address these challenges, we explored a variety of strategies. On the architectural side, we increased the model

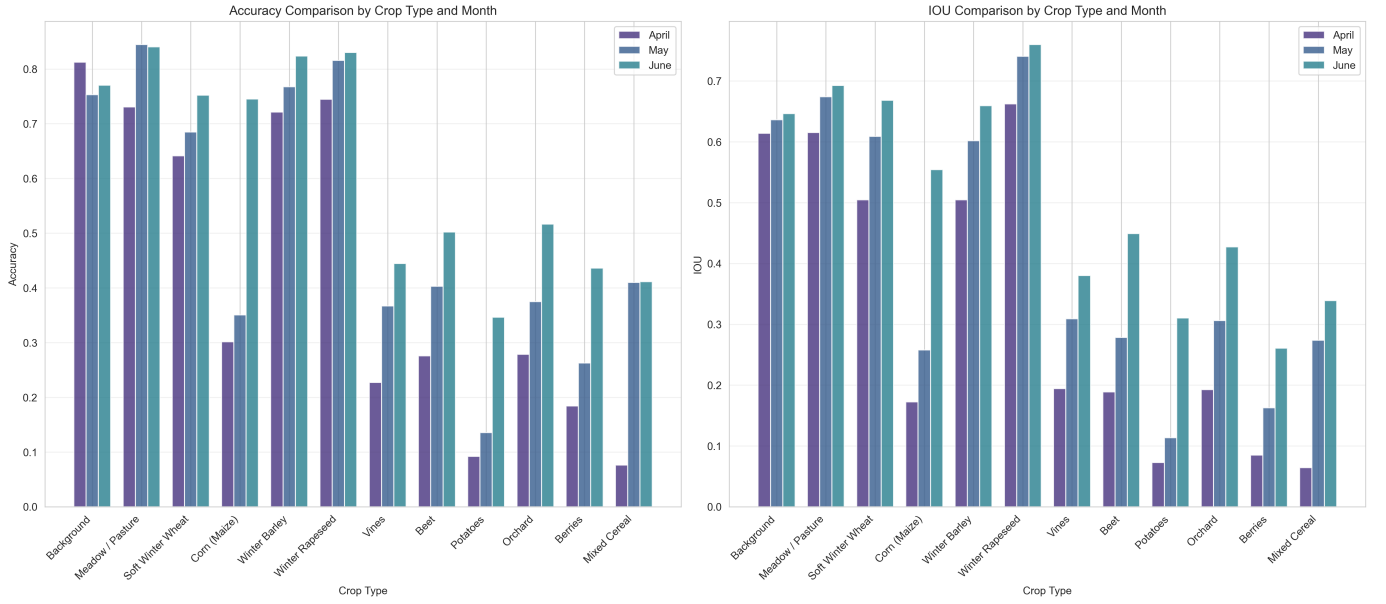


Fig. 11

TABLE IX: Aggregation map CropMapsSwitzerland

New Classes	Original Classes
Background / Non-crop	Background, Special cultures, Hedge, Multiple, Forest, Non agriculture, Waters, Gardens
Meadow / Pasture	Meadow, Pasture
Soft Winter Wheat	EinkornWheat, SummerWheat, Winter-Wheat
Corn (Maize)	Maize
Winter Barley	WinterBarley
Winter Rapeseed	WinterRapeseed
Spring Barley	SummerBarley
Sunflower	Sunflowers
Vines	Vines, Hops
Beet	Sugar beets, Beets
Winter Triticale	Wheat
Fruits, Vegetables, Flowers	Pumpkin, Tobacco, Vegetables, Chicory
Potatoes	Potatoes
Leguminous Fodder	Field bean, Peas, Lupine, MixedCrop, Legumes
Soybeans	Soy
Orchard	Apples, Pears, StoneFruit, TreeCrop, Chestnut
Berries	Berries
Mixed Cereal	Oat, Grain, Rye, Spelt, Buckwheat
Sorghum	Sorghum
Oilseeds	SummerRapeseed, Linen, Hemp, Mustard
Non-field Perennials	Biodiversity encouragement area, Fallow
Void label	Void label

Mapping from original 53 Swiss crop classes to new 22 aggregated classes.

size to better accommodate the larger number of class tokens and experimented with pixel-level patching. In parallel, we tested class rebalancing and alternative loss functions. However, none of these approaches closed the performance gap to PASTIS. These results suggest that simple tweaks to the existing transformer pipeline are insufficient. Instead, substantial changes in either preprocessing (e.g., aggregation

TABLE X: PASTIS results vs aggregated classes of CropMapsSwitzerland

Metrics	PASTIS		CropMapsSwitzerland	
	Accuracy	IOU	Accuracy	IOU
Micro	0.808	0.678	0.792	0.655
Macro	0.725	0.601	0.594	0.472

Aggregated class setting for CropMapsSwitzerland vs PASTIS. While the micro-level metrics are nearly on par with the PASTIS scores, the macro-level metrics still lag behind. This gap can be attributed to the higher heterogeneity of Swiss fields, which is particularly pronounced in the smaller, less represented classes.

strategies, hierarchical labeling, or improved sampling of minority classes) or model design (e.g., hierarchical architectures, spatial-context integration, or multimodal fusion) will likely be required.

Alongside these baseline experiments, we conducted several targeted analyses. Incorporating environmental variables such as temperature provided initial insights into the potential value of auxiliary data in transformers. Midseason classification experiments demonstrated how crop phenology drives separability, establishing a baseline for possible real world applications for the public sector. Finally, experiments with class aggregation showed that grouping similar crops leads to a significant performance boost, pointing towards hierarchical classification as a promising future direction.

Taken together, this study establishes a solid basis for future work on CropMapsSwitzerland. It highlights both the strengths of transformer-based methods and the specific points where they struggle in a realistic, national-scale

setting. By pinpointing these challenges, our analysis lays the groundwork for methodological advances that can better handle class imbalance, heterogeneity, and fine-grained crop taxonomies — ultimately moving closer to robust and operational crop mapping in complex agricultural landscapes.

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